TRR356 workshop · 10 November 2026

Hands-on session

Bacterial phylogenomics: pangenomes, gain and loss, gene-content GWAS

Repository and handout: github.com/HaimAshk/bacterial-phylogenomics-workshop

How this session works

Pick a track

The toy example, the *Pseudomonas* results, or your own genomes.

Go at your own pace

Follow the handout in any order. Everything is in the repository.

Ask at any time

Unmute, write in the chat, or share your screen when something does not run.

Handout: step-by-step commands, as PDF and Markdown in the **handout/** folder of the repository.

Repository: github.com/HaimAshk/bacterial-phylogenomics-workshop

Set up: step 0 of the handout

```
git clone https://github.com/HaimAshk/bacterial-phylogenomics-workshop.git
cd bacterial-phylogenomics-workshop
mamba env create -f envs/phylo-genomes.yml -p ./conda_envs/phylo-genomes
mamba env create -f envs/phylo-gwas.yml -p ./conda_envs/phylo-gwas
mkdir -p work && cd work
cp ../data/pseudomonas/* ../data/toy_example/* ../scripts/* .
```

- Run from the repository folder; the environments take a few minutes.
- **phylo-gwas**: GLOOME, treeWAS, pyseer. **phylo-genomes**: skani, PopPUNK, dRep.
- Activate with **conda activate ./conda_envs/phylo-gwas**.

What you have to work with

Microbializer
orthogroups, presence/absence matrix, species tree

→

GLOOME
gains, losses, rates, homoplasy per family

→

treeWAS, pyseer
gene families linked to a trait

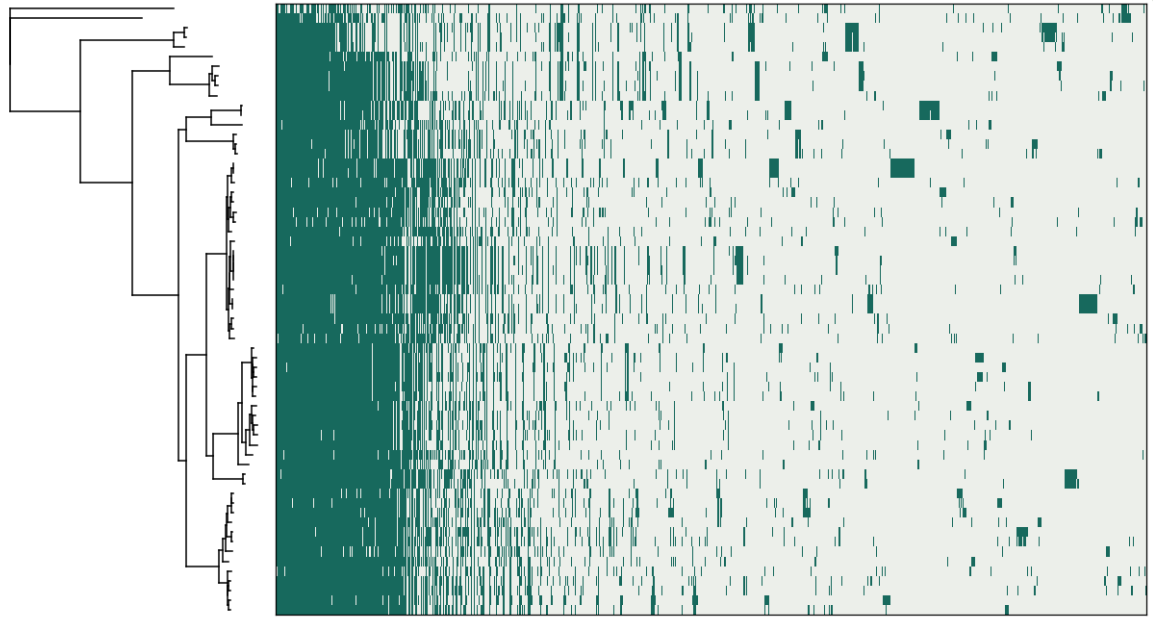
→

PanGene-O-Meter
gene-content distances and clusters

Two datasets: the **toy example** (32 genomes, 60 families, known answer) and ***Pseudomonas syringae*** (63 genomes, 26,703 families).

Files: data/toy_example/ and data/pseudomonas/ in the repository

Explore the *Pseudomonas* results



Tree and matrix: made in step 4

Run **phylo_plots.py**, then open **plots/tree_matrix.png**: blocks that follow clades, and patterns that do not.

Gains and losses: made in step 5

Branches.tsv and **ParsimonyAndHomoplasyScores.tsv** from the GLOOME run.

Gene content vs ANI: made in step 4

In **plots/gcs_vs_ani.png** (same run as above): similar ANI, very different gene content.

Left: presence/absence of gene families along the species tree (63 genomes). Plots come from phylo_plots.py (handout step 4).

Things to explore

Homoplasy

Which families are scattered across the tree, and which follow the clades?

Gains and losses

On which branches do gains and losses cluster?

Toy example

Which method finds the planted families, and which hits are decoys? Compare with toy2_truth.tsv.

Genome pairs

Which pairs have high ANI but very different gene content?

Your own data: put your genomes in **fna/** and follow handout steps 1 to 5.

If something does not run

1. Run the command with **--help**: options change between versions.
2. Copy the **error message** into the chat, not a description of it.
3. Or share your screen and we look at it together.

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